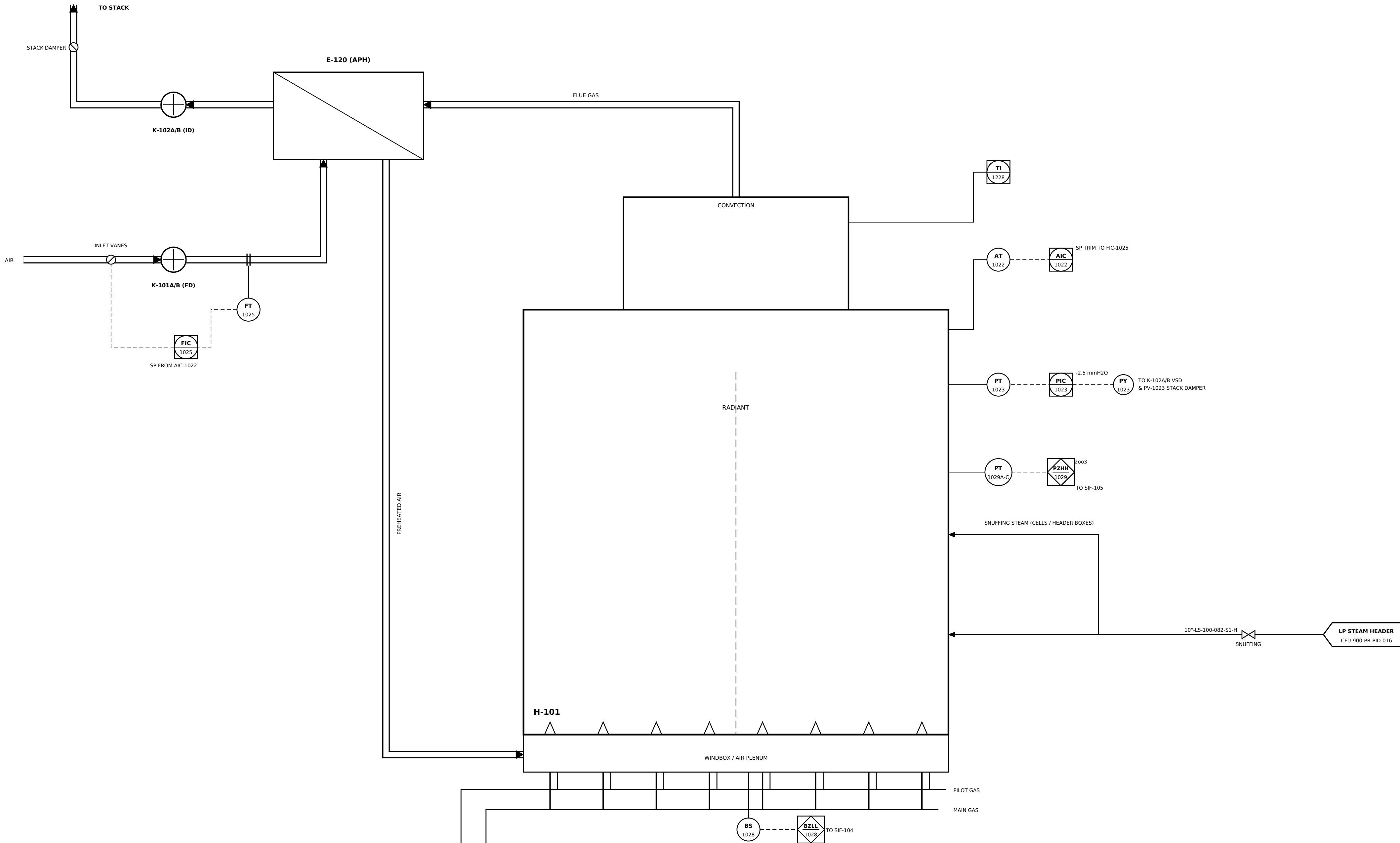


H-101
Atmospheric crude charge heater
60.7 MW abs. / 67.4 MW fired; 2-cell cabin, 8 passes, 144 rad.
tubes 6" x 18.3 m
DESIGN 38.5 barg / 545 C; CA 3 mm
DUTY 60.68 MW
MAT'L Tubes A335 P9 (9Cr-1Mo); convection P5 / CS studded

E-120
H-101 cast-iron/glass-tube air preheater
Flue gas 380 -> 160 C
DESIGN None barg / 450 C
MAT'L Cast iron / borosilicate

K-101A/B
H-101 forced-draft fan
96 t/h flue gas
MOTOR 90 kW

K-102A/B
H-101 induced-draft fan
96 t/h flue gas
MOTOR 200 kW

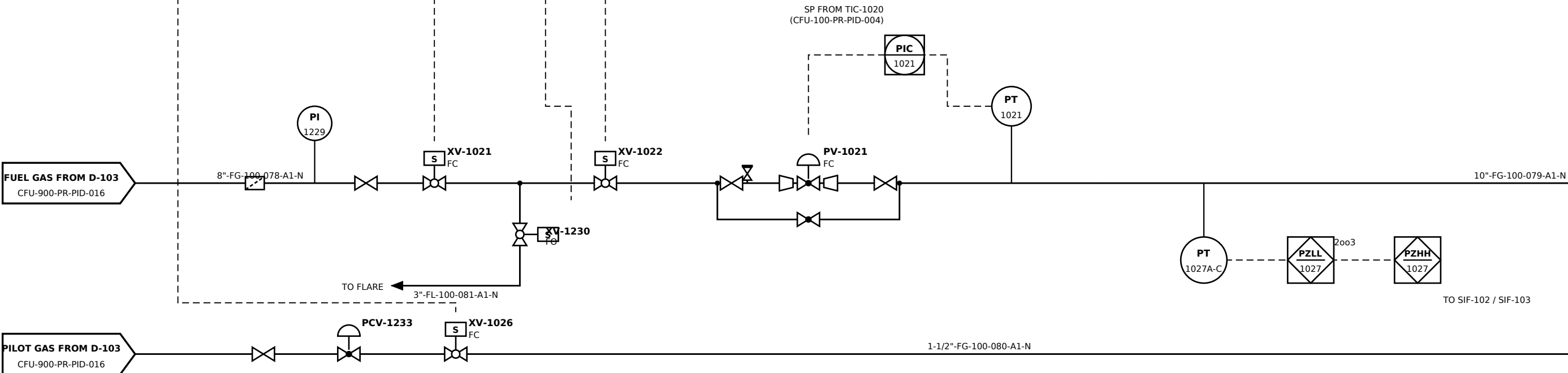


INPUTS: FZLL-1011, 1018, TZHH-1020 (CFU-100-PR-PID-004), PZLL/PZHH-1027, BZLL-1028, PZHH-1029 (THIS SHEET)

H-101 BMS / SIS (SIL-RATED PLC)

- SIF-101 (SIL 2)**
H-101 low-low pass flow (any pass) -> trip fuel
- SIF-102 (SIL 1)**
H-101 low-low fuel gas pressure -> trip fuel
- SIF-103 (SIL 1)**
H-101 high-high fuel gas pressure -> trip fuel
- SIF-104 (SIL 2)**
H-101 loss of flame (all burners) -> trip fuel
- SIF-105 (SIL 1)**
H-101 high-high arch pressure / loss of ID fan
- SIF-106 (SIL 1)**
H-101 high-high COT -> trip fuel

TRIP: XV-1021/1022 + XV-1026 CLOSE, VENT XV OPEN



NOTES

- Burner management per API 556 / NFPA 85; purge, light-off and trip sequences in BMS (SIS logic).
- Main fuel SSOVs XV-1021/1022 with vent XV (double block & bleed); all fail closed (vent fails open).
- Fuel / air cross-limiting; AIC-1022 trims FIC-1025 set point; PZLL-1027 low-select vs. minimum fire.
- Snuffing steam manifold valves located >= 15 m from heater (API 560); one per cell + header boxes.
- Ducts shown double-line; K-101A/B and K-102A/B are 2 x 100 % (one fan symbol shown per pair).
- Line numbers: <NPS> -<fluid> -<area> -<seq> -<class> -<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
- All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
- Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM ATMOSPHERIC HEATER H-101 - FIRING & AIR SYSTEM			
DISCIPLINE		STAGE	SCALE
PROCESS		FEED	NTS
		SHEET	
		6 OF 17	
DRAWN		CHECKED	APPROVED
CLAUDE CODE		-	-
		DATE	
		2026-10-02	
DRAWING NUMBER			REV
CFU-100-PR-PID-005			A